

Data Sources: ISRO Lunar Datasets & Portals

Chandrayaan-2: The DFSAR (Dual-Frequency Synthetic Aperture Radar) data and derived products are archived at the ISRO Science Data Archive (ISSDC) and delivered via the PRADAN portal. According to ISRO, the full-polarimetric L-/S-band radar captured ~1400 scenes to create **north/south polar mosaics (80–90° latitude)** [6†L39-L47] [32†L146-L154] . These “Level-3C” products – including CPR, polarimetric decomposition (Odd/Even/Volume/Helix), etc. – are freely available on ISSDC’s PRADAN interface [6†L39-L47] [32†L171-L179] . (A PRADAN login is required; data are PDS4-formatted [34†L133-L141] with XML labels.) CH2 data releases have included DFSAR calibration (starting Aug 2021) and polar mosaics (Nov 2025) [6†L49-L57] .

Chandrayaan-1 Mini-SAR (Mini-RF): The polarimetric CPR data from Mini-SAR (Mini-RF, 2.38 GHz, 75 m resolution) are archived in NASA’s PDS (Geosciences node). The PDS Geosciences listing notes that **Mini-RF data (raw, calibrated, maps, mosaics)** are available [18†L249-L258] . In fact, the Level-3 polar CPR mosaics (north/south poles above 80°) were delivered as PDS3 image files (32-bit float, polar stereographic) [47†L19-L27] [47†L33-L42] . These PDS bundles (e.g. *ch1-orb-mini-rf*) are browseable and downloadable via NASA’s PDS (no login). ISRO’s ISDA also maintains Chandrayaan-1 archives [18†L243-L251] , but NASA’s PDS is the primary source for Mini-SAR data (including CPR mosaics and underlying Level-1/2 SAR data).

Other ISRO Lunar Data: Chandrayaan-1 carried additional imagers (e.g. TMC, HySI, M3) but CPR is only from radar. M3 (Moon Mineralogy Mapper) data exist on NASA PDS (Cartography node) [18†L249-L258] . Chandrayaan-3 has no radar orbiter (APXS/ChACE-3 on lander only). In summary, **CPR is only from Mini-SAR (CH1) and DFSAR (CH2)**, while DOP (degree of polarization) is a derived quantity from CH2’s full-pol data [43†L29-L37] . No ISRO dataset provides DOP directly; it must be computed from DFSAR Stokes/Coherency parameters (see *Processing Pipeline* below).

Key References: The ISRO press release explicitly states the polar mosaic products (CPR, etc.) are on ISSDC/PRADAN [6†L49-L57] . NASA’s PDS confirms Mini-RF archiving [18†L249-L258] . A recent ISRO/PRL article defines the CPR and DOP criteria used for CH2 (CPR>1 and DOP<0.13) [43†L29-L37] .

Polarimetric Products: CPR & DOP Content

- **Circular Polarization Ratio (CPR):** Both Mini-SAR (CH1) and DFSAR (CH2) products include CPR. CH1 Level-3 mosaics are single-band CPR images (no DOP), produced by ratio of same- vs. opposite-circular channels [47†L19-L27] . CH2 Level-3C mosaics include a CPR band (derived from the full-pol data) [6†L39-L47] . In ISSDC metadata/labels, CPR is typically a floating-point band. For example, CH1 Mini-RF PDS3 labels define $CPR = SC/OS$ (same/opp circular power) [47†L19-L27] and store it in an 8107×8107 image (75 m/pix, polar stereographic) [47†L33-L42] . CH2 DFSAR CPR values similarly range ~0→1 (unitless).
- **Degree of Polarization (DOP):** CH2’s full polarimetry enables computing DOP (fraction of power in polarized component) [43†L29-L37] . DOP is not directly provided in ISRO data products; it must be calculated from the coherency or Stokes matrix (e.g. using PolSARPro or custom code) [43†L29-L37] . In the CPR/DOP framework, PRL notes that volumetric scattering (ice) yields high CPR (>1) and low DOP (<0.13) [43†L29-L37] . Practically, one computes $DOP = (MaxPolarizedPower)/(TotalPower)$

from the covariance matrix. (Exact formula depends on definition – e.g. see Sinha *et al.* 2026 [43†L29-L37] .)

- **Data Formats & Encodings:** All ISRO lunar datasets use PDS standards. Chandrayaan-2 DFSAR files are PDS4 (with `.xml` labels) – GDAL and `pds4_tools` can read them directly. Chandrayaan-1 Mini-RF was originally PDS3 (`.img/.lbl`), but a new PDS4 bundle exists [18†L243-L251] . Some level-3 products may also be mirrored as GeoTIFFs for convenience, but authoritative downloads are in PDS (binary or GeoTIFF inside PDS) format [36†L212-L221] [47†L19-L27] . Calibration tables or instrument files may be HDF5 (rare in ISRO context) or plain text.

[6†L39-L47] [32†L146-L154] *Figure: (Left) High-res optical mosaic and (Right) DFSAR polarimetric image of Peary Crater (lunar north pole). The DFSAR image is a false-color composite of polarimetric scattering (green=volume, red=odd-bounce, blue=even-bounce) as delivered in ISRO's Level-3 products [6†L39-L47] . The optical view (left) is shown for context.*

Extraction Pipeline: Tools & Steps

1. Data Download:

2. **ISSDC/PRADAN:** Register at pradan.issdc.gov.in to access Chandrayaan-2 DFSAR products. Use the “Browse and Download” interface or MapBrowse tool. Data comes as zipped PDS4 bundles or file trees. Example: DFSAR polar mosaics are found under payload “SAR” (Level-3C products) on the CH2 page [6†L49-L57] . Downloads may be multi-GB; use `wget` or Python `requests` (session cookie after login) to batch-download.
3. **NASA PDS:** For Chandrayaan-1 Mini-RF, access [PDS Geosciences Node](http://pds-geosciences.wustl.edu). The *CHANDRAYAAN-1 Mini-RF* bundle (PDS4) contains CPR mosaics [18†L249-L258] . You can download via web or FTP (`pdsgeorv1.wustl.edu`). No login needed. Example command:

```
wget -r -np -nd -A "*.img;*.xml;*.lbl" https://pds-geosciences.wustl.edu/lunar/ch1-orb-1-mrffr-1-pdr-v1/ch1mrf_0xxx/data/
```

This grabs the CPR mosaics (files like `*.img`) and labels. (Adjust path to target `mosaics/` or `data/sar/` subfolders.)

4. Parsing & Inspection:

5. **PDS Labels:** Use [pds4_tools](#) or GDAL to inspect metadata. For PDS4:

```
from pds4_tools import pds4_read
ds = pds4_read("path/to/DFSAR_Level3C_bundle.xml")
print(ds.descriptors) # list datasets
```

For PDS3 (Mini-RF): simply open `.img` with GDAL. For example, GDAL can read PDS3 via:

```
gdalinfo fsb_XXXXX_3cp_pju_90n000_v1.lbl
```

Check the georeference (polar stereographic center, radii ~1737.4 km 【47†L33-L42】). Verify CENTER_LATITUDE , pixel scale (MAP_SCALE=0.075 km) 【47†L37-L42】 .

6. Data Reading:

7. **GDAL/Rasterio:** Use GDAL (via command-line or Python GDAL/Rasterio) to load the imagery:

```
import rasterio
dataset = rasterio.open("fsb_XXXXX_3cp_pju_90n000_v1.img")
cpr_band = dataset.read(1) # CPR values
meta = dataset.meta
```

For DFSAR PDS4: GDAL supports PDS4 natively 【36†L212-L221】 , so simply:

```
ds = rasterio.open("DFSAR_polar_mosaic_level3C.img.xml")
cpr = ds.read(1)
```

8. **pds4_tools:** Can also extract band data:

```
root = pds4_read("dfsar_L3C_bundle.xml")
image = root.image['Data_Quality'].get_data() # example band
```

9. **Compute DOP (if needed):** If only raw Stokes/coh matrix is available (Level-2), calculate DOP. For example, using PolSARPro or numpy:

```
# Simplified: assume covariance matrix elements (S1,S2,S3) per pixel
# DOP = sqrt(1 - (4*det(Kocorr) / trace(Kocorr)^2))
```

Details depend on input. Tools like the [polsarpro](#) library or custom code using numpy can compute DOP.

10. Reprojection to Lunar Coordinates:

11. The data are in *moon-centered polar stereographic*. To work in latitude/longitude (planetocentric), use PROJ string like +proj=stere +lat_0=90 +lat_ts=90 +lon_0=0 +a=1737400 +b=1737400 +units=m +no_defs +R=1737400 . For example:

```
gdalwarp -t_srs "PROJCS['Moon_IAU', ...]" dfsar_mosaic.img dfsar_latlon.tif
```

(GDAL knows IAU2000:MOON if PROJ is updated.) In Python/rasterio:

```
from pyproj import CRS
moon_crs = CRS.from_proj4("+proj=latlong +a=1737400 +b=1737400")
dst = rasterio.warp.reproject(src, src_crs=ds.crs, dst_crs=moon_crs)
```

12. Ensure angular coordinates use planetocentric lat/long. The PDS labels often specify planetocentric latitude `【47†L37-L42】` .

13. Resampling & Gridding (if needed):

14. If combining data at different resolutions, use `gdalwarp` or `rasterio` resampling (e.g. bilinear). For instance:

```
gdalwarp -tr 0.025 0.025 -r bilinear dfsar_latlon.tif dfsar_25m.tif
```

15. To sample point values, create a grid of points (e.g. every 25 m or at some spacing) covering the mosaic region. Or simply iterate pixels as points.

16. Per-Point Sampling & Extraction:

17. Use GDAL or Python to loop over pixels or shapefile points. E.g., with Python:

```
import numpy as np, pandas as pd
data = []
for i in range(ds.height):
    for j in range(ds.width):
        lon, lat = ds.xy(i,j, offset='center', reverse=False)
        val_cpr = cpr[j,i]
        # Compute DOP here if needed from other bands...
        data.append((lon, lat, val_cpr))
df = pd.DataFrame(data, columns=['lon', 'lat', 'CPR'])
df.to_csv("polar_cpr_points.csv", index=False)
```

18. For large images, use block-read or GDAL's `ReadAsArray` and vectorized extraction.

19. Filtering & Sorting:

20. Once in tabular form (CSV/Parquet), use pandas or SQL to filter (e.g. `df[df.CPR>1]` for high-CPR spots) or spatial queries (within lat/long bounds).

Mermaid Pipeline:

```
graph LR
    A[Register & Download Data] --> B[Parse Metadata (PDS4)]
    B --> C[Load Raster (GDAL/Python)]
    C --> D[Compute CPR/DOP (if needed)]
    D --> E[Reproject & Resample]
    E --> F[Sample Points (lat,lon)]
    F --> G[Save to CSV/DB]
    G --> H[Filter/Analyze]
```

Data Volumes, Access & Sizes

- **Chandrayaan-2 DFSAR:** ~1400 scenes $\approx$ few TB raw; Level-2 products ~100s of GB; Level-3 polar mosaics are smaller (one per pole per parameter). For example, each CPR mosaic image (north or south) at 25 m spacing may be ~3–5 GB (assuming ~12000×12000 pixels). ISRO reported releasing “polar mosaics” [\[6†L49-L57\]](#) ; expect several GB each. Data are public with registration; no payroll. Download speeds are modest (HTTP), so use bulk tools or ISSDC bulk APIs.
- **Chandrayaan-1 Mini-SAR:** The polar mosaic files (PDS3 .img) are ~250 MB each [\[45†L10-L18\]](#) . Underlying SAR scenes (Level-1) are larger but fewer. Total Mini-SAR archive is a few GB. Access via NASA PDS is open (FTP or HTTP).
- **Authentication:** ISSDC requires signup but is free for academic use. NASA PDS requires none. Expect to create an account at [ISSDC](#) to download CH2.

Validation & Metadata Checks

- **Metadata Fields:** Always verify PDS labels: check `TARGET_NAME=MOON` , `COORDINATE_SYSTEM` , map projection, and scaling factors [\[47†L17-L25\]](#) [\[47†L33-L42\]](#) . Ensure coordinate center (north/south pole) matches product. Confirm date/time stamps and instrument mode in the label (e.g. `MODE_ID`) to ensure correct data subset.
- **Data Integrity:** Use provided checksums or `volume_descriptors` to verify file integrity. For PDS4 bundles, run `pds4_validate` . For mosaics, compare known maps: e.g., the published CPR mosaic (ISRO image) [\[6†L60-L68\]](#) can be visually compared. CPR values should lie in plausible range (0–2). Inconsistencies (NaNs, out-of-range) may indicate errors.
- **Cross-Reference:** Compare against published results or auxiliary data. For instance, compare sample CPR values at known icy craters (e.g. Peary, Faustini) with literature (e.g. Sinha *et al.*). Check consistency between north/south products at overlap boundaries (80° edge).

Database & Storage Recommendations

Format	Pros	Cons
PostGIS	Full SQL querying, spatial indexing, scalable, ACID; good for filtered queries (e.g. CPR>1). Can index lat/lon. Supports geometry (points).	Overkill for extremely large (billions) of points; needs a server; ingest time.
Parquet (with Geoparquet)	Columnar, efficient storage and compression, easily partitioned; works well with Python/pandas or Spark; supports filtering (Snappy, predicate pushdown).	Less native geospatial query (but arrow/Parquet + GeoParquet can store coords). Not as easy interactive as DB.
GeoPackage	Single-file SQLite with spatial extensions; moderate size; portable; supports attributes and geometries natively; works in GIS.	Slower for very large tables; 2GB file size limit (in practice); not distributed.
Flat CSV/JSON	Simple, readable; easy to generate; universally accessible.	Very large files (hundreds of millions of rows) become unwieldy; no spatial indexing; slow filtering.
Tile Server/ MBTiles	If tiling into map tiles (e.g. for web display), MBTiles stores raster/tiles; great for visualization.	Not suitable for attribute queries (CPR values not directly queryable); best for serving images.

Recommendation: For “point cloud” of polar CPR/DOP, a spatial database like **PostGIS** or a columnar format like **Parquet (GeoParquet)** is ideal. PostGIS excels at complex spatial queries (e.g. “find all points within 1km of crater X with CPR>1.2”), while Parquet is efficient for batch analytics and large-scale partitioning (e.g. by latitude band). For large-scale filtering/sorting on many attributes, Parquet combined with tools like DuckDB or Spark is effective. A compromise could be to store raw points in Parquet (for bulk processing) and load critical subsets into PostGIS for interactive queries.

Sample Script (Python)

```
import rasterio, numpy as np, pandas as pd
from pds4_tools import pds4_read

# 1. Open a Chandrayaan-2 DFSAR Level-3 CPR product (PDS4 XML label)
root = pds4_read("DFSAR_LEVEL3C_NCPR.xml") # example file name
img = root.image['CircularPolarizationRatio'].get_data() # PDS4 image
# 2. Reproject (if needed)
with rasterio.open("DFSAR_N_CPR.img") as src:
    cpr = src.read(1) # numpy array
```

```

transform = src.transform
crs = src.crs
# 3. Compute lat/lon for each pixel
rows, cols = cpr.shape
xs = np.arange(cols)
ys = np.arange(rows)
lons, lats = rasterio.transform.xy(transform, ys, xs, offset='center')
# (Use meshgrid for full grid, or sample points of interest)
# 4. Flatten and build DataFrame
lon_flat = np.repeat(lons, rows)
lat_flat = np.tile(lats, cols)
cpr_flat = cpr.flatten()
df = pd.DataFrame({'lon': lon_flat, 'lat': lat_flat, 'CPR': cpr_flat})
# 5. Filter out missing (if any) and save
df = df[np.isfinite(df.CPR)]
df.to_csv("dfsar_NCPR_points.csv", index=False)

```

CLI Example: Reproject with GDAL and extract stats:

```

# Reproject CPR mosaic to geographic coords
gdalwarp -t_srs "EPSG:4326" DFSAR_N_CPR.img DFSAR_N_CPR_latlon.tif
# Inspect min/max values
gdalinfo -mm DFSAR_N_CPR_latlon.tif

```

【6†L39-L47】 【47†L19-L27】 *Table: The Chandrayaan-2 DFSAR Level-3 polar mosaics (north/south) include CPR, decompositions (Odd/Even/Volume/Helix) 【6†L39-L47】 . The Chandrayaan-1 Mini-SAR Level-3 mosaics similarly provide CPR as a single-band raster 【47†L19-L27】 【47†L33-L42】 .*